

D.SURYA VIDYADHAR

Hardware engineer (6+ Exp)

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ABOUT ME

Electronics Hardware Engineer with 6+ years of industrial experience in designing and testing electronic systems. Skilled in embedded hardware design, troubleshooting, and working with automotive and aerospace-grade electronics. Experienced in managing hardware projects from concept to production, ensuring quality and compliance. Strong collaborator focused on delivering reliable and efficient solutions. Committed to continuous learning and applying new technologies in engineering.

WORK EXPERIENCE

BluJ AeroSpace Private Limited - Hyderabad- Hardware design engineer

MAY 2025 - NOW

- Designed 16KW ESC(control & power board)

Lakshmi Electrical Control Systems Ltd - Coimbatore -Hardware Engineer Smart meter dept

JUNE 2024 - MAY 2025

- Worked 1 phase and 3phase smart meters for house hold application

Sona Comstar Automotive (Chennai) - Hybrid EV R&D as Hardware engineer Employed through DNCL Technologies, Bangalore, serving as the payroll company

JUNE 2023 - JUNE 2024

- Gained automotive experience working on 16 kW and 1.2 kW motor controllers.

Free lancer - For more than four years, led freelance engagements with a focus on innovation and client satisfaction, establishing a reputation for reliable and efficient independent work.

2018 - 2023

Concentrix - Hardware engineer

OCT 2015 - FEB 2017

- Hands-on experience in developing and implementing IoT products for house hold applications.

Madammal technologies - Hardware engineer

SEP 2012 - FEB 2015

- Experience in building and mentoring academic projects for graduation students

SKILL

Designed schematics for 8-bit, 16-bit, 32-bit, and 32-bit TriCore microcontrollers. Prepared bills of materials and developed embedded software, managing the entire process from requirements to production and commercial deployment, Experience in performing worst case analysis (Monte Carlo and Extreme Value Analysis)

Experience in design practices as below

- Requirements and scope analysis. Electrical, control electronics and hardware module requirement capturing. Defining the hardware design requirements
- Selection of components based on Automotive & aerospace component standards
- Design calculation and simulation, Derating, EOL
- Schematic design and BOM release
- PCB Layout guidelines
- Board bring up
- Module testing and reports
- Experienced in performing schematic worst-case analysis, including Monte Carlo simulations and extreme value analysis, to ensure robust and reliable circuit performance under varying conditions.
- Updating all design changes next version
- Board debugging and Integration of Hardware & board bring up.
- Testing motors on Dyno level for linearity and accuracy
- Worked with CAN tool while doing dyno test
- Worked in test Jig environment while developing EV controllers

Micro Controllers Familiar with

- Texas Instruments: MSP430FR5969, TMS320F28379D
- NXP- S32K Automotive (ASIL D) with SBC
- Infineon tricore microcontroller
- AI Thinker: ESP 8266-12f

Design & simulations tools Familiar with

- Altium Designer
- OrCAD Capture
- LTspice
- Tina-TI

ACHIEVEMENTS

Secured 1st place in Fight Coronavirus (COVID-19) using IoT and ML Challenge for the project "Fight COVID-19 Bolt quarantine survival robot". (This is national wide challenge is conducted by BOLT-IOT Bangalore)

- Feel free to watch the video. <https://www.youtube.com/watch?v=Y9TS5N6g5ds&t=17s>
- Certificate https://drive.google.com/file/d/1a1ZZd1-EqEPdMYrtNBogaCP_whL4YEY8/view?usp=sharing
- Project published on BOLT IOT website click below link.
<https://www.boltiot.com/projects/fight-covid-19-quarantine-robot-using-bolt-iot>

BEST PAPER NOMINEE in Texas Instruments MCU DESIGN CHALLENGE 2014 for "Mobile based password authentication system".

- Feel free to watch the video. <https://www.youtube.com/watch?v=bAOonpT1oTY&t=13s>

PROFESSIONAL EXPERTISE:

16KW Motor ESC

Description: Developed 16KW Motor ESC for eVTOL, This was designed for sensorless Field-Oriented Control (FOC) technology with SiC MOSFETs. The system design is based on a 3-phase inverter architecture. Its functionality is to rotate the motor using the sensorless FOC technique

- Schematic design and BOM release
- PCB Layout guidelines
- Board bring up
- Software integration while validating the PCBA

SMART ENERGY METER

Description: Developed single phase smart energy meter current rating 5-30A & 10-60A which supports 90 day of data logger.

- Schematic design and BOM release
- PCB Layout guidelines
- Board bring up
- Software integration while validating the PCBA

EV Motor controller 8 to 15KW

Description: Developed EV motor controller by using Asciil-D safety level.

Responsibilities:

- Bench marking
- Selection of components
- Individual module testing
- Performed Design calculation and simulation
- Schematic design and BOM release
- PCB Layout guidelines
- Board bring up
- Module testing and report

EV Motor controller 1.2KW

Description: Developed 1.2 kw motor controller for fully active suspension for four wheeler by using Ascil-B safety features

Responsibilities:

- design calculation and simulation, Derating, EOL
- Schematic design and BOM release
- performed worst case analysis
- PCB Layout guidelines
- Board bring up
- Motor unit assembly constraints
- Dyno testing using CAN
- Performed DFMEA
- Resolving testing issues.

Data Processing Unit

Description: This project will act as Data Processing Machine by connecting Weight machine and Milk analyzer and will proceed received data as an output form on GLCD, Mini Thermal printer, USB,RTC,. DPU have days and monthly wise. When we need take print out. DPU will operates battery and power supply.

Responsibilities:

Requirements and scope analysis. Electrical, control electronics and hardware module requirement capturing. Defining the hardware design requirements

Architecting hardware solutions and defining functional requirement

- Selection of components, ICs.
- DE-Rating Analysis
- Hardware schematics design & implementation
- Battery charging circuit design
- GLCD,5*5 KEYPAD, I2C, USB, SPI, RTC, Buzzer, LED
- Creation of EBOM
- PCB review, Gerber review
- Board debugging and Integration of Hardware & board bring up.

Atmega 8 Evaluation board

MC : Atmega8

Tools : Orcad Capture(cadence tool),AVR studio.

Role and Responsibilities: developed schematic on Orcad Capture, developed test codes for development board

EDUCATION

- Bachelors - B.Tech-ECE in 2012 at BVC College of Engineering, Andhra Pradesh
- Intermediate(+2) - in 2007 at ADITYA JUNIOR COLLEGE,MANDAPETA, Andhra Pradesh
- SSC - in 2005 at GOVT HIGH SCHOOL,MANDAPETA, Andhra pradesh

TRAININGS ATTENDED:

- Attended Texas Instruments Design day - Bangalore
- Internship Program (45 days)on Advanced Robotic in ROBOSAPIENS - NOIDA
- Internship Program on PCB Designing in ROBOSAPIENS(45 days) - NOIDA
- Participated in ROBOSAPIENS workshop on MOBI-BOTRICKS at AVANTI COLLEGE OF ENGG.
- Participated in ROBO-OPUS workshop on e-Robotics at BVCEC
- Participated a seminar on EMBEDDED SYSTEMS at SKBR COLLEGE Amalapuram.
- Participated in Robotics work shop conducted by Li2 innovations at JNTU Kakinada

PERSONAL DETAILS

Father's Name : satyanarayana donthamsetti
Marital Status : Married
Date of Birth : 30-08-1990
Languages Known : English, Hindi, and Telugu
Native Place : Mandapeta, Andhra pradesh

Declaration

I hereby declare that the information furnished above is true to the best of my knowledge.